Hints: Limits – 2. Recurrences solvable in closed form

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Let $f(x) = (x^2 - 9)/8$. Its fixed points are 9 and -1; note $f'(9) = \frac{9}{4}$ (repelling) and $f'(-1) = -\frac{1}{4}$ (attracting). Also f is even and $f(x) \geqslant -\frac{9}{8}$, so only $|x_1|$ matters and $x_n \geqslant -\frac{9}{8}$ for $n \geqslant 2$.

Hint 2. Use $f(x) - 9 = \frac{(x-9)(x+9)}{8}$: if $|x_1| > 9$ then $x_2 > 9$ and $x_n - 9$ grows at least geometrically, so $x_n \to +\infty$; if $|x_1| = 9$ then $x_n = 9$ for $n \geqslant 2$.

Hint 3. If $|x_1| < 9$ then $x_2 \in [-\frac{9}{8}, 9)$. Use $f(x) + 1 = \frac{(x+1)(x-1)}{8}$: for $x \in [-\frac{9}{8}, 9)$ we have $|x - 1| < 8$, so $d_n = |x_n + 1|$ decreases; then all later x_n lie in $[-\frac{9}{8}, -1 + d_2]$, where $|x - 1| \leqslant 8q$ for a fixed $q < 1$, so $d_n \to 0$ geometrically.

Hint 4. Answer: $\lim x_n = -1$ if $|x_1| < 9$, $= 9$ if $|x_1| = 9$, $= +\infty$ if $|x_1| > 9$.

Problem 2

Hint 1. Uniqueness and existence: $g(x) = x - \varepsilon \sin x$ is continuous with $g'(x) = 1 - \varepsilon \cos x \geqslant 1 - |\varepsilon| > 0$, and $g(x) \to \pm\infty$ as $x \to \pm\infty$.

Hint 2. Rewrite the equation as $x = t + \varepsilon \sin x$ and iterate $x_{n+1} = t + \varepsilon \sin x_n$ from any x_0. Since $|\sin u - \sin v| \leqslant |u - v|$, the map is a contraction with constant $|\varepsilon| < 1$: $|x_{n+1} - x| \leqslant |\varepsilon|\,|x_n - x|$, so $x_n \to x$ geometrically (this is Kepler's equation).

Problem 3

Hint 1. Solve the recurrence: the characteristic roots are 2 and $\frac{1}{2}$; find $F(n)$ in closed form.

Hint 2. $F(n) = 2^n - 2^{-n}$, so with $y_n = 2^{2^n}$ the terms are $\dfrac{1}{y_n - 1/y_n} = \dfrac{y_n}{y_n^2 - 1}$, and $y_{n+1} = y_n^2$.

Hint 3. $\dfrac{y}{y^2 - 1} = \dfrac{1}{y - 1} - \dfrac{1}{y^2 - 1}$, so the series telescopes: $\sum_{n \geqslant 0} \frac{1}{F(2^n)} = \frac{1}{y_0 - 1} = 1$. Rational.

Problem 4

Hint 1. Compute: $1, 2, 6, 22, 90, 394, \ldots$ These are the (large) Schröder numbers, which satisfy the integral recurrence $s_n = s_{n-1} + \sum_{k=0}^{n-1} s_k s_{n-1-k}$, i.e. their generating function $S(x)$ satisfies $S = 1 + xS + xS^2$. The task is to prove $a_n = s_n$.

Hint 2. Translate the given recurrence into an equation for $f(x) = \sum a_n x^n$: multiplying by x^{n+2} and summing gives the first-order linear ODE

$$x(1 - 6x + x^2) f'(x) + (1 - 3x) f(x) = 1 + x.$$

A formal power series solution of this ODE is unique (the coefficient of x^m determines a_m), so it suffices to check that S satisfies it.

Hint 3. From $S = 1 + xS + xS^2$ get $S(x) = \dfrac{1 - x - \sqrt{1 - 6x + x^2}}{2x}$; differentiate and verify the ODE. Then $a_n = s_n$, which are integers by the convolution recurrence.

Problem 5

Hint 1. Use the generating function $f(x) = \sum a_n x^n$ (by induction $0 < a_n \leqslant 1$, so it converges for $|x| < 1$). The required limit is $f(\frac{1}{2})$.

Hint 2. The recurrence says $(n+1)a_{n+1} = \sum_{k=0}^{n} a_k \cdot \frac{1}{n-k+2}$, i.e. $f'(x) = f(x)g(x)$ with $g(x) = \sum_{m\geqslant 0} \frac{x^m}{m+2} = \dfrac{-\ln(1-x) - x}{x^2}$.

Hint 3. An antiderivative of g vanishing at 0 is $1 + \frac{(1-x)\ln(1-x)}{x}$, so $f(x) = e\,(1-x)^{(1-x)/x}$ and the limit is $f(\frac{1}{2}) = \dfrac{e}{2}$.

Problem 6

Hint 1. Decompose into partial fractions: $\dfrac{1}{(4k+1)(4k+2)(4k+3)(4k+4)} = \dfrac{1}{6}\Big(\dfrac{1}{4k+1} - \dfrac{3}{4k+2} + \dfrac{3}{4k+3} - \dfrac{1}{4k+4}\Big)$.

Hint 2. Write $\frac{1}{4k+j} = \int_0^1 x^{4k+j-1}\,dx$ and sum the geometric series (justify swapping, e.g. by monotone convergence on $[0,1)$): the sum is $\frac{1}{6}\int_0^1 \frac{(1-x)^3}{1-x^4}\,dx = \frac{1}{6}\int_0^1 \frac{(1-x)^2}{(1+x)(1+x^2)}\,dx$.

Hint 3. $\frac{(1-x)^2}{(1+x)(1+x^2)} = \frac{2}{1+x} - \frac{x+1}{1+x^2}$, giving the answer $\dfrac{\ln 2}{4} - \dfrac{\pi}{24}$.

Problem 7

Hint 1. The map $x \mapsto x^2 - 2$ is $t + t^{-1} \mapsto t^2 + t^{-2}$. Write $x_1 = \sqrt{5} = t + t^{-1}$.

Hint 2. $t = \frac{1+\sqrt{5}}{2}$, so $x_n = t^{2^{n-1}} + t^{-2^{n-1}}$, and the product telescopes: $(t - t^{-1})x_1 x_2 \cdots x_n = t^{2^n} - t^{-2^n}$.

Hint 3. Hence the ratio equals $\dfrac{1}{t - t^{-1}} \cdot \dfrac{t^{2^n} - t^{-2^n}}{t^{2^n} + t^{-2^n}} \to \dfrac{1}{t - t^{-1}} = 1$.

Problem 8

Hint 1. The sum is the "odd part" of a binomial expansion: $S_n = \sum_k \binom{n}{2k+1} 5^k = \dfrac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{2\sqrt{5}}$.

Hint 2. By Theorem 2, S_n satisfies $S_{n+1} = 2S_n + 4S_{n-1}$ ($1 \pm \sqrt{5}$ are the roots of $x^2 - 2x - 4$), with $S_1 = 1$, $S_2 = 2$. Prove $2^{n-1} \mid S_n$ by strong induction. (Equivalently, $S_n = 2^{n-1}F_n$.)

Problem 9

Hint 1. The number of ways is the constant term of $(x^{-1} + 1 + x)^{p-1}$. Choosing first which $2k$ minutes are jumps and then which k of them go to the right, $f(p) = \sum_k \binom{p-1}{2k}\binom{2k}{k}$ (check: $f(3) = 3$, $f(5) = 19$, $f(7) = 141$). For $p = 2$: $f(2) = 1$.

Hint 2. Modulo p one has $\binom{p-1}{j} \equiv (-1)^j$, so $f(p) \equiv \sum_{k=0}^{(p-1)/2} \binom{2k}{k} \pmod p$ for odd p. With $m = \frac{p-1}{2} \equiv -\frac{1}{2}$ one has $\binom{2k}{k} = (-4)^k\binom{-1/2}{k} \equiv (-4)^k\binom{m}{k}$ (all denominators are $k!$ with $k < p$).

Hint 3. Hence $f(p) \equiv \sum_{k=0}^{m}\binom{m}{k}(-4)^k = (-3)^{(p-1)/2}$, which by Euler's criterion and $\left(\frac{-3}{p}\right) = \left(\frac{p}{3}\right)$ is $\equiv 0, 1, -1$ according as $p = 3$, $p \equiv 1$, $p \equiv 2 \pmod 3$ (and $f(2) = 1$).

Problem 10

Hint 1. Get rid of the root: $(a_{n+2} - 5(a_{n+1} + a_n))^2 = 4(6a_{n+1}^2 + 15a_{n+1}a_n + 6a_n^2)$. Expand and look at the symmetry of the result.

Hint 2. It becomes $a_{n+2}^2 + a_{n+1}^2 + a_n^2 = 10(a_{n+2}a_{n+1} + a_{n+1}a_n + a_n a_{n+2})$, symmetric in the three consecutive terms. So for $n \geqslant 2$ both a_{n+2} and a_{n-1} are roots of $z^2 - 10(a_{n+1} + a_n)z + (a_{n+1}^2 + a_n^2 - 10a_{n+1}a_n) = 0$.

Hint 3. The sequence is increasing, so $a_{n-1} < 5(a_{n+1} + a_n) \leqslant a_{n+2}$: they are the two different roots, and Vieta gives $a_{n+2} = 10(a_{n+1} + a_n) - a_{n-1}$. With $a_3 = 71$ computed directly, integrality follows by induction.

Problem 11

(a)

Hint 1. Compute: $1, 2, 5, 14, 41, 122, \ldots$; guess $a_n = \frac{3^n+1}{2}$.

Hint 2. Verify by induction: $\frac{3^{n+1}+1}{2} \cdot \frac{3^{n-1}+1}{2} = 3^{n-1} + \left(\frac{3^n+1}{2}\right)^2$ (both sides minus $\frac{3^{2n}+1}{4}$ equal $\frac{10 \cdot 3^{n-1}}{4}$). Equivalently $a_{n+1} = 4a_n - 3a_{n-1}$, which gives integrality directly.

(b)

Hint 1. $\frac{1}{a_n} = \frac{2}{3^n+1} < 2 \cdot 3^{-n}$: compare with a geometric series. It converges.

(c)

Hint 1. By (a), $\frac{2^n}{a_{2^n}} = \frac{2^{n+1}}{y_n+1}$ with $y_n = 3^{2^n}$, and $y_{n+1} = y_n^2$. Check the identity $\dfrac{1}{y+1} = \dfrac{1}{y-1} - \dfrac{2}{y^2-1}$.

Hint 2. Hence $\dfrac{2^{n+1}}{y_n+1} = \dfrac{2^{n+1}}{y_n-1} - \dfrac{2^{n+2}}{y_{n+1}-1}$, the series telescopes and its sum is $\dfrac{2}{y_0-1} = 1$ (the remainder $\frac{2^{N+1}}{3^{2^N}-1} \to 0$). Rational.